

Zixiang Chen

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EDUCATION

- **University of California, Los Angeles.** Ph.D. candidate in Computer Science (Advisor: Quanquan Gu).
Sep.2019 - Present
- **Tsinghua University.** B.Sc in Pure and Applied Mathematics.
Aug.2015- Jul. 2019

RESEARCH INTERESTS

Foundation Models, Learning Theory, Reinforcement Learning, Interdisciplinary AI

PROJECTS

Guided Discrete Diffusion for Electronic Health Record Generation

- Interdisciplinary project with OPTUM Labs.
- Research project in 2023: Proposed a new EHR generator named [EHR-D3PM](#), a discrete diffusion process specifically tailored for EHR data; model empowered by a transformer architecture with linear attention. Experiments demonstrated that it outperforms existing generative baselines on comprehensive fidelity metrics while preserving privacy.
- Research project in 2024: Apply diffusion transformer to incorporate temporal data.

Fine-tuning Protein Diffusion Model

- Intern project at ByteDance.
- Fine-tuned the Protein Diffusion Model on the supervised SFT dataset to improve its fidelity on the test dataset while largely preserving its variety in generating different conformations.

PUBLICATIONS (* INDICATES EQUAL CONTRIBUTION)

• Conference Publications

1. Fast Sampling via Discrete Non-Markov Diffusion Models

Zixiang Chen, Huizhuo Yuan, Yongqian Li, Yiwen Kou, Junkai Zhang, Quanquan Gu

In Proc. of Advances in Neural Information Processing Systems (NeurIPS) 38, Vancouver, Canada, 2024.

2. Self-Play Fine-Tuning of Diffusion Models for Text-to-Image Generation

Huizhuo Yuan*, **Zixiang Chen***, Kaixuan Ji*, Quanquan Gu

In Proc. of Advances in Neural Information Processing Systems (NeurIPS) 38, Vancouver, Canada, 2024.

3. Matching the Statistical Query Lower Bound for k-sparse Parity Problems with Stochastic Gradient Descent
Yiwen Kou*, **Zixiang Chen***, Quanquan Gu, Sham M. Kakade
In Proc. of Advances in Neural Information Processing Systems (NeurIPS) 38, Vancouver, Canada, 2024.
4. Self-Play Fine-Tuning Converts Weak Language Models to Strong Language Models
Zixiang Chen*, Yihe Deng*, Huizhuo Yuan*, Kaixuan Ji, Quanquan Gu
In Proc. of the 41th International Conference on Machine Learning (ICML), Vienna, Austria, 2024.
5. Understanding Transferable Representation Learning and Zero-shot Transfer in CLIP
Zixiang Chen*, Yihe Deng*, Yuezhi Li, Quanquan Gu,
In Proc. of the 12th International Conference on Learning Representations (ICLR), Vienna, Austria, 2024.
6. How Many Pretraining Tasks Are Needed for In-Context Learning of Linear Regression?
Jingfeng Wu, Difan Zou, **Zixiang Chen**, Vladimir Braverman, Quanquan Gu, Peter L Bartlett
In Proc. of the 12th International Conference on Learning Representations (ICLR), Vienna, Austria, 2024. (Spotlight)
7. Implicit Bias of Gradient Descent for Two-layer ReLU and Leaky ReLU Networks,
Yiwen Kou*, **Zixiang Chen*** and Quanquan Gu
In Proc. of Advances in Neural Information Processing Systems (NeurIPS) 37, New Orleans, LA, USA, 2023.
8. Why Does Sharpness-Aware Minimization Generalize Better Than SGD?
Zixiang Chen*, Junkai Zhang*, Yiwen Kou, Xiangning Chen, Cho-Jui Hsieh and Quanquan Gu
In Proc. of Advances in Neural Information Processing Systems (NeurIPS) 37, New Orleans, LA, USA, 2023.
9. Benign Overfitting for Two-layer ReLU Networks,
Yiwen Kou*, **Zixiang Chen***, Yuezhou Chen and Quanquan Gu,
In Proc. of the 40 th International Conference on Machine Learning (ICML), Hawaii, USA, 2023.
10. Finite-Sample Analysis of Learning High-Dimensional Single ReLU Neuron,
Jingfeng Wu*, Difan Zou*, **Zixiang Chen***, Vladimir Braverman, Quanquan Gu and Sham M. Kakade,
In Proc. of the 40th International Conference on Machine Learning (ICML), Hawaii, USA, 2023.
11. A General Framework for Sample-Efficient Function Approximation in Reinforcement Learning,
Zixiang Chen*, Chris Junchi Li*, Huizhuo Yuan*, Quanquan Gu, Michael Jordan,
In Proc. of the 11th International Conference on Learning Representations (ICLR), Kigali, Rwanda, 2023. (Spotlight)
12. How Does Semi-supervised learning with Pseudo-labelers Work? A Case Study,
Yiwen Kou, **Zixiang Chen**, Yuan Cao and Quanquan Gu,
In Proc. of the 11th International Conference on Learning Representations (ICLR), Kigali, Rwanda, 2023.

13. [Understanding Train-Validation Split in Meta-Learning with Neural Networks](#),
Xinzhe Zuo, **Zixiang Chen**, Huaxiu Yao, Yuan Cao and Quanquan Gu,
In Proc. of the 10th International Conference on Learning Representations (ICLR), Kigali, Rwanda, 2023.
14. [Towards Understanding Mixture of Experts in Deep Learning](#),
Zixiang Chen, Yihe Deng, Yue Wu, Quanquan Gu, Yuanzhi Li,
Advances in Neural Information Processing Systems (NeurIPS) 36, New Orleans, LA, USA, 2022.
15. [Benign Overfitting in Two-layer Convolutional Neural Networks](#),
Yuan Cao*, **Zixiang Chen***, Mikhail Belkin, Quanquan Gu,
Advances in Neural Information Processing Systems (NeurIPS) 36, New Orleans, LA, USA, 2022 (**Oral**).
16. [Almost Optimal Algorithms for Two-player Zero-Sum Linear Mixture Markov Games](#),
Zixiang Chen, Dongruo Zhou and Quanquan Gu,
In Proc. of the 33rd International Conference on Algorithmic Learning Theory (ALT), Paris, France, 2022.
17. [Faster Perturbed Stochastic Gradient Methods for Finding Local Minima](#),
Zixiang Chen*, Dongruo Zhou* and Quanquan Gu,
In Proc. of the 33rd International Conference on Algorithmic Learning Theory (ALT), Paris, France, 2022.
18. [Self-training Converts Weak Learners to Strong Learners in Mixture Models](#),
Spencer Frei*, Difan Zou*, **Zixiang Chen*** and Quanquan Gu,
In Proc of the 23rd International Conference on Artificial Intelligence and Statistics, Valencia, Spain, 2022.
19. [How Much Over-parameterization Is Sufficient to Learn Deep ReLU Networks?](#)
Zixiang Chen*, Yuan Cao*, Difan Zou* and Quanquan Gu,
In Proc. of the 9th International Conference on Learning Representations (ICLR), 2021.
20. [A Generalized Neural Tangent Kernel Analysis for Two-layer Neural Networks](#),
Zixiang Chen, Yuan Cao, Quanquan Gu and Tong Zhang,
Advances in Neural Information Processing Systems (NeurIPS) 34, 2020.

• **Workshop Publications**

1. [Fast Sampling via De-randomization for Discrete Diffusion Models](#)
Zixiang Chen, Huizhuo Yuan, Yongqian Li, Junkai Zhang, Yiwen Kou, Quanquan Gu
NeurIPS 2023 Workshop on Diffusion Models
2. [Understanding Transferable Representation Learning and Zero-shot Transfer in CLIP](#)
Zixiang Chen*, Yihe Deng*, Yuanzhi Li, Quanquan Gu,
NeurIPS 2023 Workshop on Mathematics of Modern Machine Learning (M3L). **Oral**

• Preprints

1. Rephrase and respond: Let large language models ask better questions for themselves

Yihe Deng, Weitong Zhang, **Zixiang Chen**, Quanquan Gu

2. Stein Neural Sampler,

Tianyang Hu*, **Zixiang Chen***, Hanxi Sun*, Jincheng Bai, Mao Ye, Guang Cheng.

3. Guided Discrete Diffusion for Electronic Health Record Generation

Jun Han*, **Zixiang Chen***, Yongqian Li, Yiwen Kou, Eran Halperin, Robert E. Tillman, Quanquan Gu

4. Convergence of Score-Based Discrete Diffusion Models: A Discrete-Time Analysis

Zikun Zhang, **Zixiang Chen**, Quanquan Gu

INVITED TALKS

“Benign Overfitting in Two-layer Convolutional Neural Networks”

Math Machine Learning seminar MPI MiS + UCLA

May 2022

“Towards Understanding Mixture of Experts in Deep Learning”

Representation Learning Theory TTIC Chicago Summer Workshop

August 2022

“A General Framework for Sample-efficient Function Approximation in Reinforcement Learning”

Group meeting hosted by Lin F. Yang, Department of Electrical and Computer Engineering, UCLA

December 2022

“Why Does Sharpness-Aware Minimization Generalize Better Than SGD?”

Group meeting hosted by Deanna Needell, Department of Mathematics, UCLA

November 2023

“Self-Play Fine-Tuning Converts Weak Language Models to Strong Language Models”

Beijing Academy of Artificial Intelligence (BAAI) Qingyuan TALK

Feb 2024

NLG Seminars of Information Science Institute of USC

March 2024

RLHF reading group at Mila. Hosted by Professor Doina Precu

April 2024

Lightning talk at Google's Theory and Practice of Foundation Models Workshop

Nov 2024

PROFESSIONAL SERVICES

• Conference Reviewer

ICML, NeurIPS, ICLR, AISTAT, AAAI

• Journal Reviewer

TMLR, JMLR, JAIR

AWARDS

UCLA Dissertation Year Fellowship

June 2024

Student Travel Award for NeurIPS 2022, 2023

Scholarships in Tsinghua for Academic Excellence

Jan 2016 - 2019

Gold Medal in 30th Chinese Mathematical Olympiad

Jan 2014

TEACHING

Teaching Assistant at University of California, Los Angeles

- CS31: Intro to Computer Science (2023 Winter UCLA)
- CS 260: Machine Learning Algorithms (2022 Fall UCLA) [[Website](#)] & (2021 Fall UCLA) [[Website](#)]

LANGUAGE SKILLS

Programming: Python, R, MATLAB

Machine Learning: PyTorch, Hugging Face

Data Analysis: Pandas, NumPy, SciPy

Experiment Management: Hydra